Exercise 3

Given data set $(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, Y_{n+1}) \in \mathcal{X} \times \mathcal{Y}$ with $\mathcal{Y} = \mathbb{R}$, level $\alpha \in (0, 1)$, we discretize $[-B, B]$, where $B > 0$, by setting, for a positive integer M and all $m \in \{1, \ldots, M\}$:

$$y^{(m)} = -B + \frac{m-1}{M-1} \times 2B.$$

Put, for any $y \in \mathcal{Y}$:

$$\mathcal{D}_{n+1}^{y} = \{(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, y)\}.$$

With a real valued score function $s((x, y); \mathcal{D})$ for $(x, y) \in \mathcal{X} \times \mathcal{Y}$ and $\mathcal{D}$ is any data set containing points from $\mathcal{X} \times \mathcal{Y}$, we consider a discretized conformal prediction set defined as follows. First of all, let I be the set of all $m \in \{1, \ldots, M\}$ such that

$$s\left((X_{n+1}, y^{(m)}); \mathcal{D}_{n+1}^{y^{(m)}}\right) \leqslant \text{Quantile}\left(\left\{s\left((X_i, Y_i); \mathcal{D}_{n+1}^{y^{(m)}}\right)\right\}_{i=1}^{n}; (1-\alpha)\left(1 + \frac{1}{n}\right)\right).$$

Then, put

$$\mathcal{C}(X_{n+1}) = \bigcup_{m \in I} (y^{(m-1)}, y^{(m+1)}).$$

Construct an example of a symmetric score function s and a distribution P of the data set such that

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1})\} = 0$$

for any choice of B and M.

Solution We construct the example as follows. First of all, map the set $\mathcal{X}$ to $\mathbb{R}$ in any way one wants. From now on, we shall identify $\mathcal{X}$ with $\mathbb{R}$ and we will write for simplicity $\mathcal{X} = \mathbb{R}$. We construct it such that the pairs (X_i, Y_i) ranged over $i \in \{1, \ldots, n+1\}$ are independent identically distributed with distribution of X_i's being any fixed continuous probability measure (say, standard normal) and $Y_i = X_i$. As of the score, we simply put

$$s((x, y); \mathcal{D}) = |x - y|$$

for all (x, y) and all data set $\mathcal{D}$. Clearly, s is symmetric because it does not even depend on $\mathcal{D}$. If $i \in \{1, \ldots, n\}$, our construction yields, almost surely, that

$$s\left((X_i, Y_i); \mathcal{D}_{n+1}^{y^{(m)}}\right) = |X_i - Y_i| = 0$$

and

$$\text{Quantile}\left(\left\{s\left((X_i, Y_i); \mathcal{D}_{n+1}^{y^{(m)}}\right)\right\}_{i=1}^{n}; (1-\alpha)\left(1 + \frac{1}{n}\right)\right) = 0$$

Therefore, $I = \emptyset$ and $\mathcal{C}(X_{n+1}) = \emptyset$ for all choice of M and all choice of B. ////